

Women's Health Reset Exam 1 — Written (Short Answer) Section

Course: MED422 — Women's Health

This document has **two parts**:

1. **Part A — Question paper** with blank answer spaces (print and write on it).
 2. **Part B — Model answers**, with marks weighted as on the paper.
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Part A — Question Paper (blank)

Case 1

A 19-year-old primigravida was admitted to the Emergency Department with severe headache and reduced fetal movements. She did not discover that she is pregnant until very late and was uncertain of her last menstrual period. The first scan was at 23 weeks gestation and according to that scan she is now 37 weeks.

On her booking visit, the blood pressure was 120/68 mmHg and urine analysis negative for proteins. The booking investigations were all normal.

This morning she woke up with frontal headache which persisted despite regular paracetamol intake. She says that her vision is a bit blurred, but she cannot be more specific about this. She also reports nausea and epigastric discomfort.

Examination

- The blood pressure is 164/106 mmHg, repeated twice at 15 min intervals as 160/110 mmHg and 164/112 mmHg.
- She is apyrexial and the heart rate is 83 beats/min.
- Her face is swollen and fundus examination is normal.
- On abdominal examination she is tender in the epigastrium and beneath the right costal margin, but the uterus is soft and non-tender.
- The legs are mildly edematous and the lower limb reflexes are very brisk, with clonus.

Lab investigations

Investigation	Patient's level	Normal range
Hemoglobin	11.6 gm/dl	11 - 14 gm/dl
Platelet count	126 x10 ³ /ml	150 - 450 x10 ³ /ml

Investigation	Patient's level	Normal range
AST	150 IU/L	6 – 32 IU/L
ALT	80 IU/L	15 – 70 IU/L
Serum creatinine	0.8 mg/dl	0.7 – 0.9 mg/dl
Serum Uric acid	0.6 mmol/L	0.14 – 0.3 mmol/L
Serum Potassium	4.0 mmol/L	3.3 – 4.1 mmol/L
Urine analysis	++ protein	Negative

A CTG was performed and revealed: baseline fetal heart rate 140 beat/min, reduced beat-to-beat variability (3–5 bpm), variable decelerations, and no accelerations.

Questions

1. What is the most likely diagnosis? Justify based on the clinical and lab findings. (2 marks)

2. Mention a further investigation that should be requested. (1 mark)

3. How would you manage this case? (3 marks)

4. Mention the mechanism of action, indication and signs of magnesium sulphate toxicity. (2 marks)

5. List 3 anti-hypertensive medications that can be used in pregnancy and their mechanism of action. (2 marks)

Case 2

A 32-year-old G4P3 attended the antenatal clinic at 36 weeks gestation requesting information about postnatal family planning. Her current pregnancy was unplanned as she was only using the safe period and external ejaculation. She has 3 children, the youngest is 16 months old, and they were all delivered vaginally at term. All babies were breastfed, and she is intending to breastfeed this baby too.

She is currently normotensive, weighing 75 Kg. Her menses are regular, recurring every 30 days, of average amount and no dysmenorrhea. She is non-smoker and has no relevant medical history.

Questions

1. Mention TWO contraceptive options that could be offered to her within the time of delivery whether vaginal or cesarean? (2 marks)

2. Give her THREE other options that could be started at the end of puerperium. (3 marks)

3. The woman asks if there is a method that would be contraindicated because of breastfeeding. (3 marks)

4. If this woman told you that her menses are usually heavy with marked dysmenorrhea, which method would be most suitable for her? (2 marks)

Part B — Model Answers

Case 1 — Model Answers (10 marks)

1. What is the most likely diagnosis? Justify based on the clinical and lab findings. (2 marks)

Diagnosis: Severe pre-eclampsia at 37 weeks with features of evolving HELLP syndrome (and signs of impending eclampsia).

Justification (award for citing the criteria):

- **New-onset hypertension after 20 weeks in the severe range** — 160/110 and 164/112 mmHg on repeat, on a background of a normal booking BP of 120/68.
- **New proteinuria** — urine ++ (booking urine was negative).
- **Severe features / end-organ involvement:** persistent headache, blurred vision and epigastric/right-upper-quadrant tenderness; brisk reflexes with clonus (neurological irritability = impending eclampsia); facial oedema.
- **Laboratory evidence:** thrombocytopenia (platelets 126), raised transaminases (AST 150, ALT 80 — both > 2× upper limit) and hyperuricaemia (0.6 mmol/L) → the low platelets + high liver enzymes suggest evolving HELLP.
- **Fetal compromise:** pathological CTG (reduced variability, variable decelerations, no accelerations) and reduced fetal movements.

2. Mention a further investigation that should be requested. (1 mark)

Any one relevant test to confirm HELLP / assess maternal status, e.g.:

- **Peripheral blood film + LDH + bilirubin** — to confirm haemolysis and complete the HELLP picture (*best single answer*).
- **Coagulation profile** (PT, aPTT, fibrinogen) — to detect DIC.
- 24-hour urinary protein or spot protein:creatinine ratio to quantify proteinuria; group and save.

3. How would you manage this case? (3 marks)

A senior-led, deliver-the-baby plan:

1. **Admit and stabilise (ABC)** — involve senior obstetrician, anaesthetist and neonatal team; secure IV access, monitor BP, urine output, reflexes and continuous CTG.
2. **Control blood pressure** — IV labetalol (or IV hydralazine, or oral nifedipine) aiming for $< 150/100$ mmHg to prevent maternal stroke.
3. **Seizure prophylaxis** — IV **magnesium sulphate** (4 g loading dose over 5–10 min, then 1 g/hour maintenance) because she has impending eclampsia (headache, visual symptoms, clonus).
4. **Deliver the baby — the definitive treatment.** At 37 weeks with severe pre-eclampsia/evolving HELLP and a pathological CTG, expedite delivery; given the non-reassuring CTG, caesarean section is likely (mode depends on cervix and fetal condition).
5. **Supportive care** — strict fluid balance, monitor for pulmonary oedema and DIC, continue magnesium and BP control for ≥ 24 hours postpartum. (Antenatal steroids are not needed for lung maturity at 37 weeks.)

4. Mention the mechanism of action, indication and signs of magnesium sulphate toxicity. (2 marks)

- **Mechanism of action:** magnesium is a membrane stabiliser that acts as a calcium antagonist and blocks NMDA receptors, reducing neuronal excitability and causing cerebral vasodilatation → prevents and controls seizures.
- **Indications:** seizure prophylaxis in severe pre-eclampsia; treatment of eclamptic seizures; fetal neuroprotection in threatened preterm birth < 32 weeks.
- **Signs of toxicity (progressive):** loss of deep tendon (patellar) reflexes first → respiratory depression → slurred speech, flushing, muscle weakness, reduced urine output → cardiac arrhythmia/arrest at high levels. **Antidote: IV calcium gluconate.**

5. List 3 anti-hypertensive medications that can be used in pregnancy and their mechanism of action. (2 marks)

Any three, with mechanism:

- **Labetalol** — combined α - and β -adrenergic blocker (lowers peripheral vascular resistance and heart rate).
- **Methyldopa** — central α_2 -adrenergic agonist (reduces central sympathetic outflow).
- **Nifedipine** — calcium channel blocker (arterial smooth-muscle relaxation / vasodilatation).
- (*Hydralazine — direct arteriolar vasodilator, used IV for acute severe hypertension.*)

Case 2 — Model Answers (10 marks)

1. Mention TWO contraceptive options that could be offered to her within the time of delivery whether vaginal or cesarean? (2 marks)

Any two long-acting/permanent methods that can be placed at the time of delivery:

- **Immediate postpartum intrauterine device** — copper IUD or LNG-IUS, inserted within 10 minutes of placental delivery at vaginal birth, or at caesarean section.
- **Bilateral tubal ligation (female sterilization)** — performed at caesarean section or as immediate postpartum mini-laparotomy (she is a para-3+ with completed family — counsel and consent).
- (*Progestogen-only implant — safe to insert immediately postpartum and compatible with breastfeeding.*)

2. Give her THREE other options that could be started at the end of puerperium. (3 marks)

Any three methods appropriate from ~6 weeks postpartum:

- **Progestogen-only pill (POP)** — safe during breastfeeding.
- **Progestogen injectable (DMPA)** — depot progestogen, breastfeeding-compatible.
- **Progestogen-only implant** (if not already placed).
- **Barrier methods** — male/female condoms or diaphragm.
- **Combined hormonal contraception** — only becomes acceptable from ≥ 6 weeks in a breastfeeding woman (avoid before 6 weeks).

3. The woman asks if there is a method that would be contraindicated because of breastfeeding. (3 marks)

- **a. Contraindicated method:** combined hormonal contraception (the **combined oral contraceptive pill** / any oestrogen-containing method).
- **b. Why:** the **oestrogen** component reduces breast-milk volume and can impair lactation, and there is an added early-postpartum venous thromboembolism risk.
- **c. For how long:** it is contraindicated for the **first 6 weeks** postpartum in a breastfeeding woman (UKMEC 4 before 6 weeks; UKMEC 2 from 6 weeks to 6 months) — so avoid it at least until 6 weeks, ideally while breastfeeding is being established.

4. If this woman told you that her menses are usually heavy with marked dysmenorrhea, which method would be most suitable for her? (2 marks)

The levonorgestrel intrauterine system (LNG-IUS / Mirena).

It is the most suitable because, besides providing effective long-acting contraception, it markedly **reduces menstrual blood loss** and **relieves dysmenorrhoea** — treating her heavy, painful periods at the same time.